```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv('/content/WA_Fn-UseC_-Telco-Customer-Churn.csv')

print("Dataset Shape:", df.shape)
print("\nFirst 5 Rows:")
display(df.head())

print("\nDataset Information:")
df.info()

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:")
print(df.duplicated().sum())

df = df.drop_duplicates()

df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')

df['TotalCharges'] = df['TotalCharges'].fillna(df['TotalCharges'].median())

print("\nMissing Values After Cleaning:")
print(df.isnull().sum())

print("\nChurn Distribution:")
print(df['Churn'].value_counts())

plt.figure(figsize=(6,4))
sns.countplot(data=df, x='Churn')
plt.title('Customer Churn Distribution')
plt.show()

print("\nGender Distribution:")
print(df['gender'].value_counts())

plt.figure(figsize=(6,4))
sns.countplot(data=df, x='gender')
plt.title('Gender Distribution')
plt.show()

print("\nContract Type Distribution:")
print(df['Contract'].value_counts())

plt.figure(figsize=(7,4))
sns.countplot(data=df, x='Contract')
plt.title('Contract Type Distribution')
plt.xticks(rotation=15)
```

```
plt.show()

print("\nPayment Method Distribution:")
print(df['PaymentMethod'].value_counts())

plt.figure(figsize=(9,4))
sns.countplot(data=df, x='PaymentMethod')
plt.title('Payment Method Distribution')
plt.xticks(rotation=45)
plt.show()

numerical_features = ['MonthlyCharges', 'TotalCharges', 'tenure']

for col in numerical_features:
    plt.figure(figsize=(7,4))
    sns.histplot(data=df, x=col, kde=True)
    plt.title(col + ' Histogram')
    plt.show()

for col in numerical_features:
    plt.figure(figsize=(7,4))
    sns.boxplot(data=df, x=col)
    plt.title(col + ' Boxplot')
    plt.show()

print("\nOutlier Detection Using IQR:")

for col in numerical_features:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]

    print("\nFeature:", col)
    print("Q1:", Q1)
    print("Q3:", Q3)
    print("IQR:", IQR)
    print("Lower Bound:", lower_bound)
    print("Upper Bound:", upper_bound)
    print("Number of Outliers:", len(outliers))

plt.figure(figsize=(7,5))
sns.countplot(data=df, x='Contract', hue='Churn')
plt.title('Contract Type and Churn')
plt.xticks(rotation=15)
plt.show()

plt.figure(figsize=(7,5))
sns.countplot(data=df, x='InternetService', hue='Churn')
plt.title('Internet Service and Churn')
plt.show()
```

```
plt.figure(figsize=(7,5))
sns.boxplot(data=df, x='Churn', y='MonthlyCharges')
plt.title('Monthly Charges and Churn')
plt.show()

plt.figure(figsize=(7,5))
sns.boxplot(data=df, x='Churn', y='tenure')
plt.title('Tenure and Churn')
plt.show()

contract_churn = pd.crosstab(df['Contract'], df['Churn'], normalize='index') * 100

print("\nChurn Percentage by Contract Type:")
print(contract_churn)

highest_contract = contract_churn['Yes'].idxmax()
print("\nContract Type with Highest Churn:", highest_contract)

payment_churn = df[df['Churn'] == 'Yes']['PaymentMethod'].value_counts()

print("\nPayment Method Used by Churned Customers:")
print(payment_churn)

print("\nMost Frequently Used Payment Method Among Churned Customers:",
      payment_churn.idxmax())

tenure_churn = df.groupby('Churn')['tenure'].mean()

print("\nAverage Tenure by Churn:")
print(tenure_churn)

if tenure_churn['Yes'] < tenure_churn['No']:
    print("\nCustomers with shorter tenure tend to churn more often.")
else:
    print("\nCustomers with shorter tenure do not have a lower average tenure among churned customers.")

group_monthly = df.groupby('Churn')['MonthlyCharges'].mean()

print("\nAverage Monthly Charges by Churn:")
print(group_monthly)

print("\nCustomer Group with Highest Monthly Charges:",
      group_monthly.idxmax())

internet_churn = pd.crosstab(
    df['InternetService'],
    df['Churn'],
    normalize='index'
) * 100

print("\nChurn Percentage by Internet Service:")
print(internet_churn)
```

```
highest_internet = internet_churn['Yes'].idxmax()

print("\nInternet Service Category with Highest Churn:",
      highest_internet)

churn_numeric = df['Churn'].map({'Yes': 1, 'No': 0})

monthly_correlation = df['MonthlyCharges'].corr(churn_numeric)
tenure_correlation = df['tenure'].corr(churn_numeric)
totalcharges_correlation = df['TotalCharges'].corr(churn_numeric)

print("\nCorrelation with Churn:")
print("MonthlyCharges:", monthly_correlation)
print("Tenure:", tenure_correlation)
print("TotalCharges:", totalcharges_correlation)

plt.figure(figsize=(8,5))
sns.boxplot(data=df, x='Churn', y='MonthlyCharges')
plt.title('Monthly Charges vs Churn')
plt.show()

plt.figure(figsize=(8,5))
sns.scatterplot(data=df, x='tenure', y='MonthlyCharges', hue='Churn')
plt.title('Tenure vs Monthly Charges by Churn')
plt.show()

features = [
    'tenure',
    'MonthlyCharges',
    'TotalCharges'
]

correlation_data = df[features].copy()
correlation_data['Churn'] = churn_numeric

correlation_matrix = correlation_data.corr()

print("\nCorrelation Matrix:")
print(correlation_matrix)

plt.figure(figsize=(8,6))
sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)
plt.title('Correlation Heatmap')
plt.show()

X = df.drop(columns=['Churn'])
y = df['Churn']

X = pd.get_dummies(X, drop_first=True)
```

```
    p  g  _       ( ,    p_          )

y = y.map({'Yes': 1, 'No': 0})

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("\nTraining Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)

processed_df = X.copy()
processed_df['Churn'] = y

processed_df.to_csv(
    'CustomerChurn_Preprocessed.csv',
    index=False
)

files.download('CustomerChurn_Preprocessed.csv')
```

```
Dataset Shape: (7043, 21)

First 5 Rows:
```

	cus D	ge	Sen zen	Pa	Dep s	te	Pho ce	Mul nes	Inte rvic	Onl rity	.	Devi ecti	Tec rt	Str TV	Stre ovie	Co	Pape illi	Pay hod	Mon rge:	Tot es	Cl
0	759 VH	Fe	0	Ye	No	1	No	No ser	DSL	No	.	No	No	No	No	Mo mo	Yes	Ele che	29.8	29.	N
1	557 GN	Ma	0	Nc	No	34	Yes	No	DSL	Yes	.	Yes	No	No	No	On	No	Mai che	56.9	188	N
2	366 QP	Ma	0	Nc	No	2	Yes	No	DSL	Yes	.	No	No	No	No	Mo mo	Yes	Mai che	53.8	108	Y
3	779 CF	Ma	0	Nc	No	45	No	No ser	DSL	Yes	.	Yes	Yes	No	No	On	No	Ban tra (au)	42.3	184	N
4	923 HQ	Fe	0	Nc	No	2	Yes	No	Fibe opti	No	.	No	No	No	No	Mo mo	Yes	Ele che	70.7	151	Y

```
5 rows × 21 columns
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

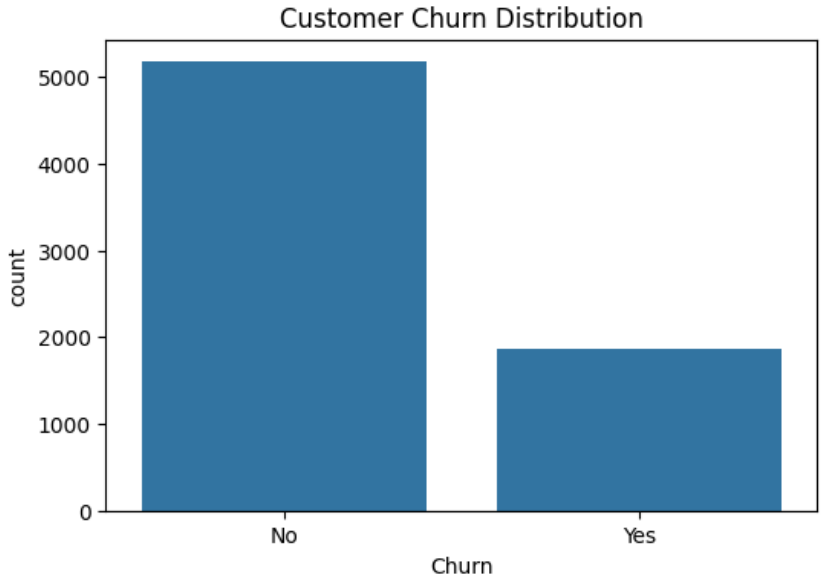
Missing Values:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
```

```
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64

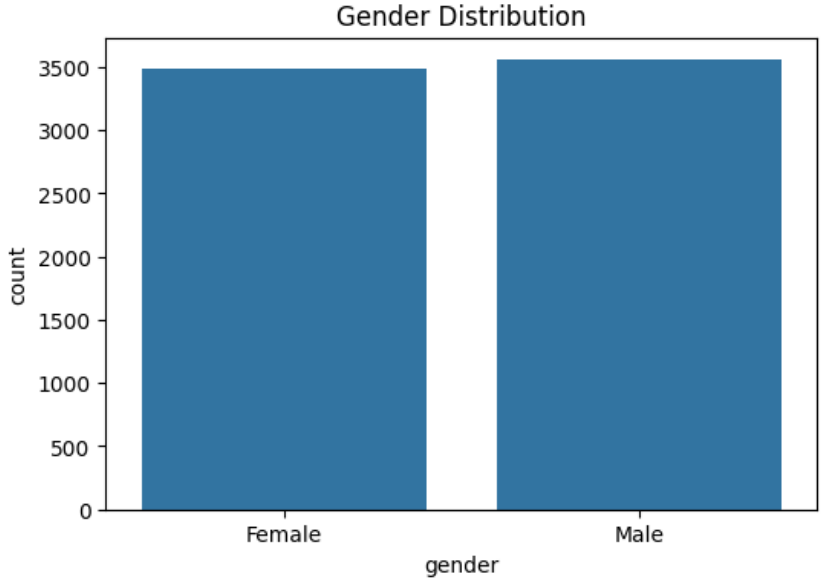
Duplicate Rows:
0

Missing Values After Cleaning:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64

Churn Distribution:
Churn
No     5174
Yes    1869
Name: count, dtype: int64
```

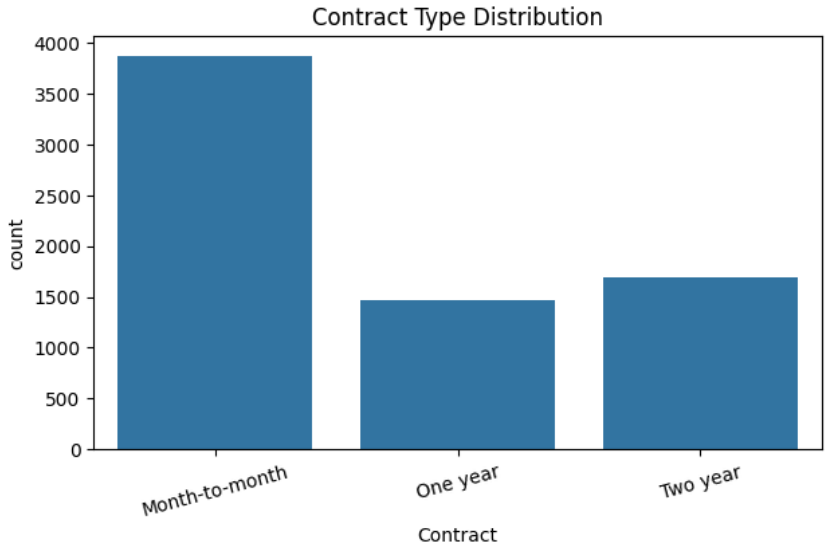



```
Gender Distribution:
gender
Male      3555
Female    3488
Name: count, dtype: int64
```

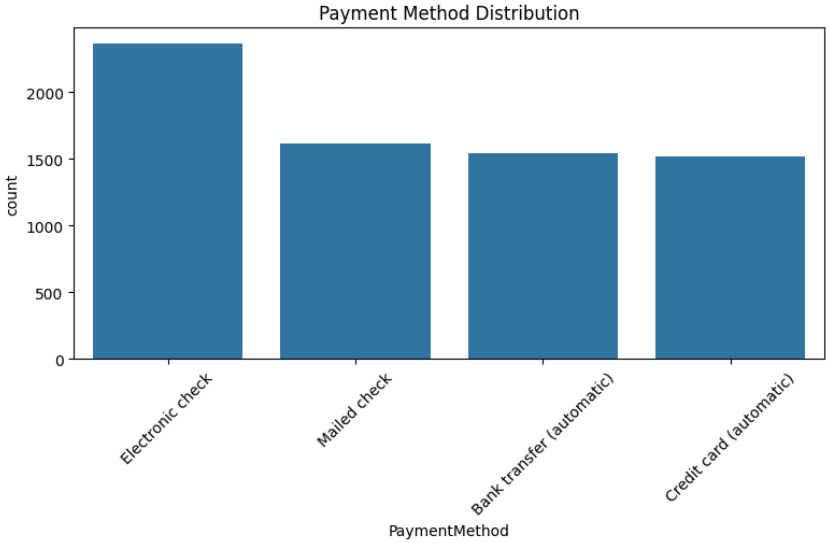


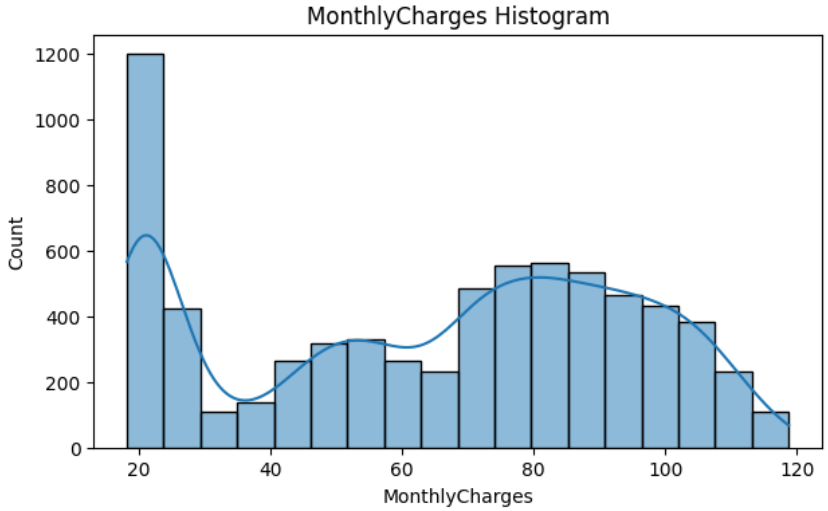

```
Contract Type Distribution:
Contract
Month-to-month    3875
Two year          1695
```

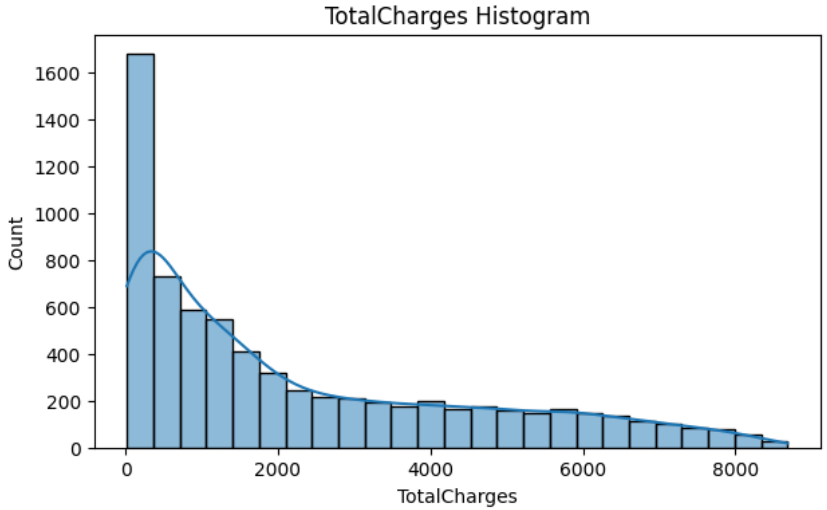
```
One year          1473
Name: count, dtype: int64
```

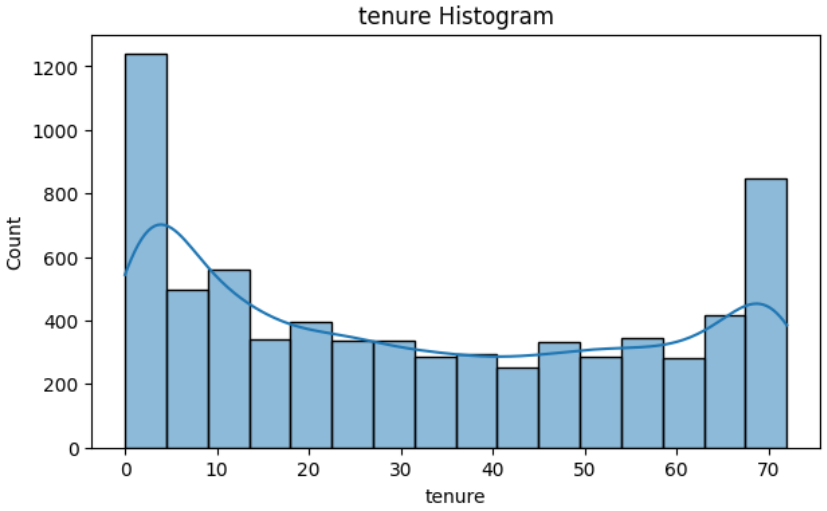


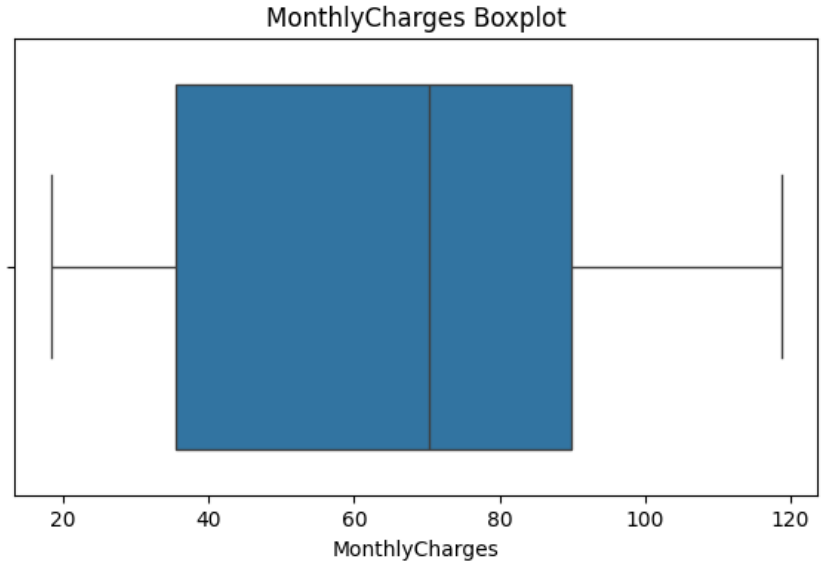

```
Payment Method Distribution:
PaymentMethod
Electronic check             2365
Mailed check                 1612
Bank transfer (automatic)    1544
Credit card (automatic)      1522
Name: count, dtype: int64
```

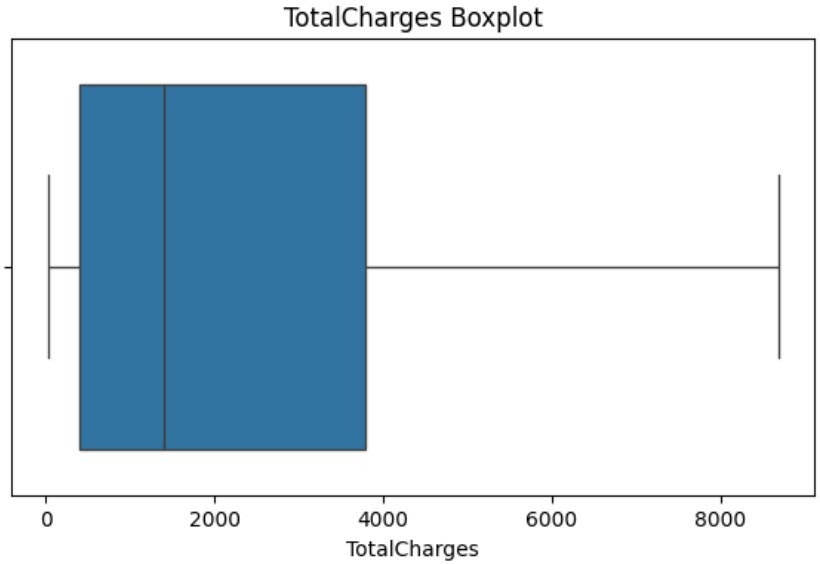


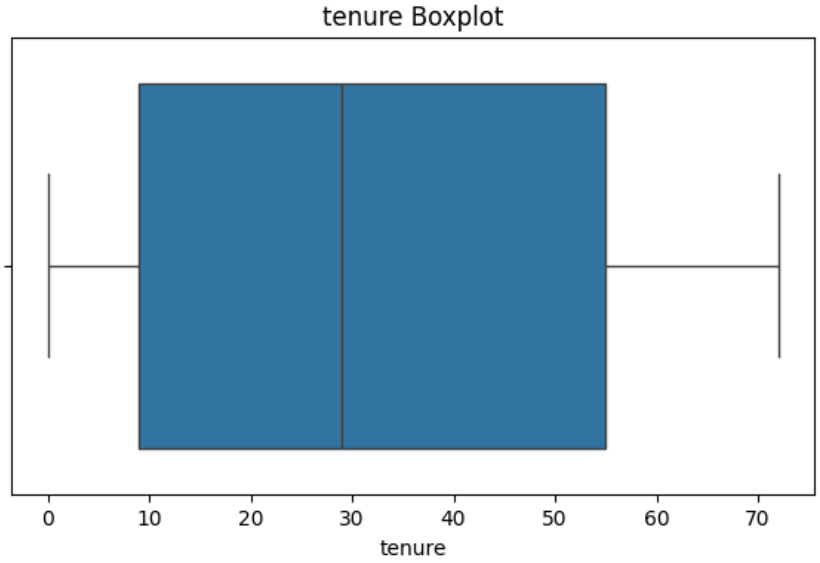

MonthlyCharges Histogram
Count
MonthlyCharges
0
200
400
600
800
1000
1200
20
40
60
80
100
120


TotalCharges Histogram
Count
TotalCharges
0
200
400
600
800
1000
1200
1400
1600
0
2000
4000
6000
8000


tenure Histogram
Count
tenure
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70


MonthlyCharges Boxplot
MonthlyCharges
20
40
60
80
100
120





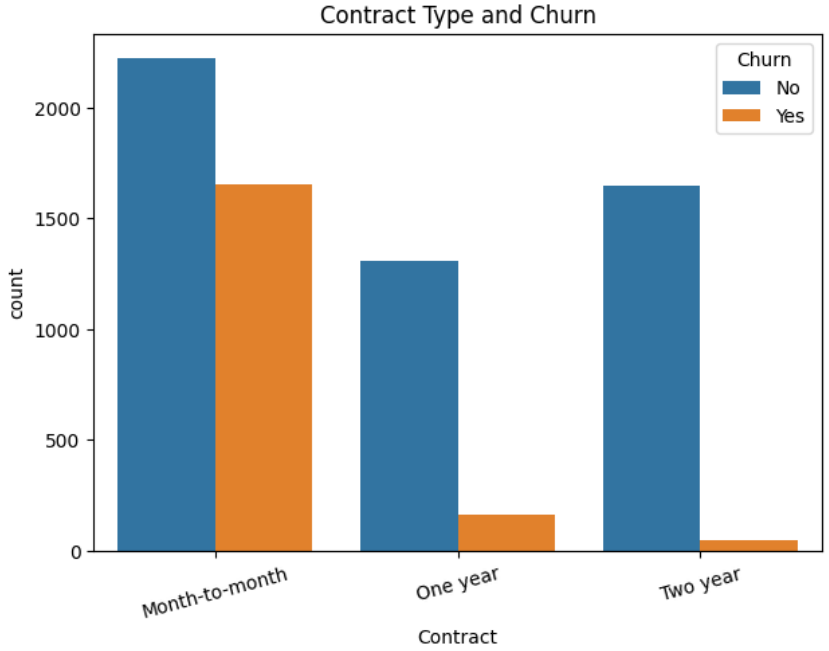

```
Outlier Detection Using IQR:

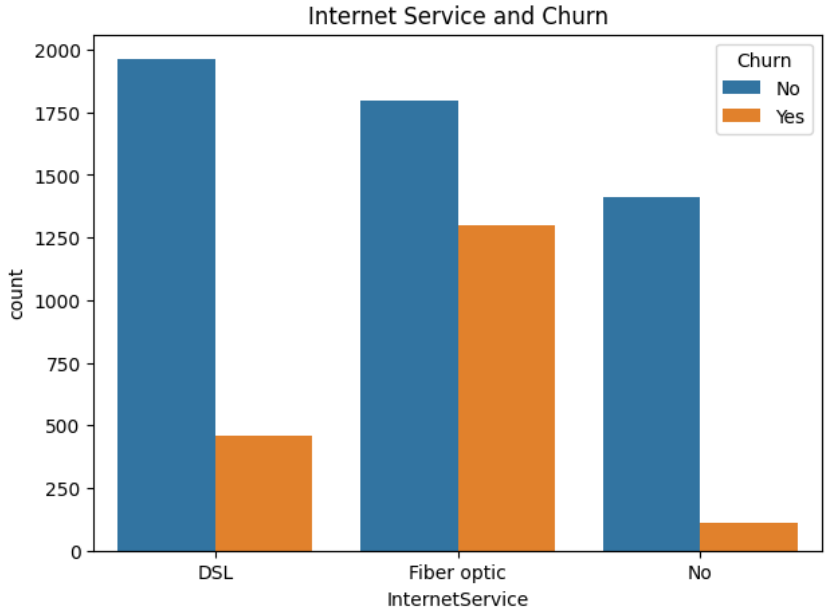
Feature: MonthlyCharges
Q1: 35.5
Q3: 89.85
IQR: 54.349999999999994
Lower Bound: -46.02499999999999
Upper Bound: 171.375
Number of Outliers: 0

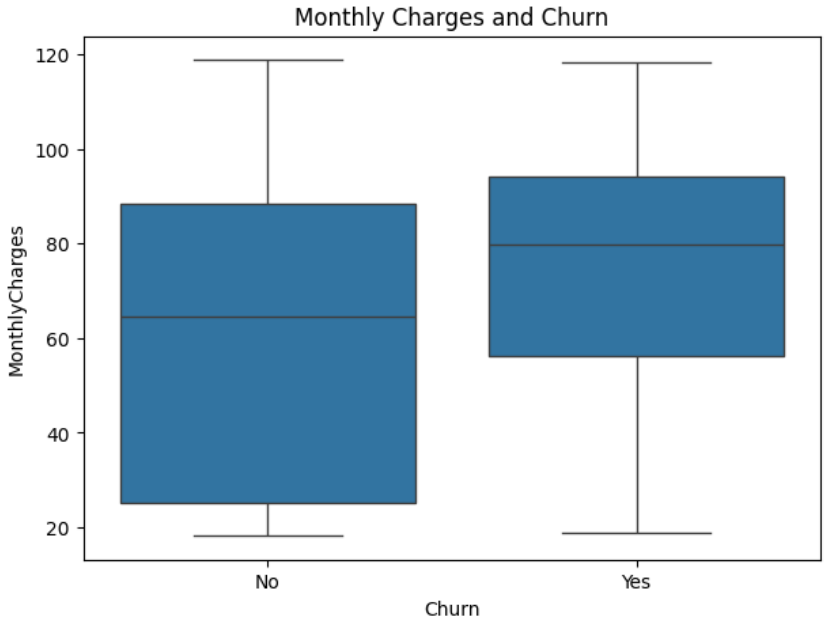
Feature: TotalCharges
```

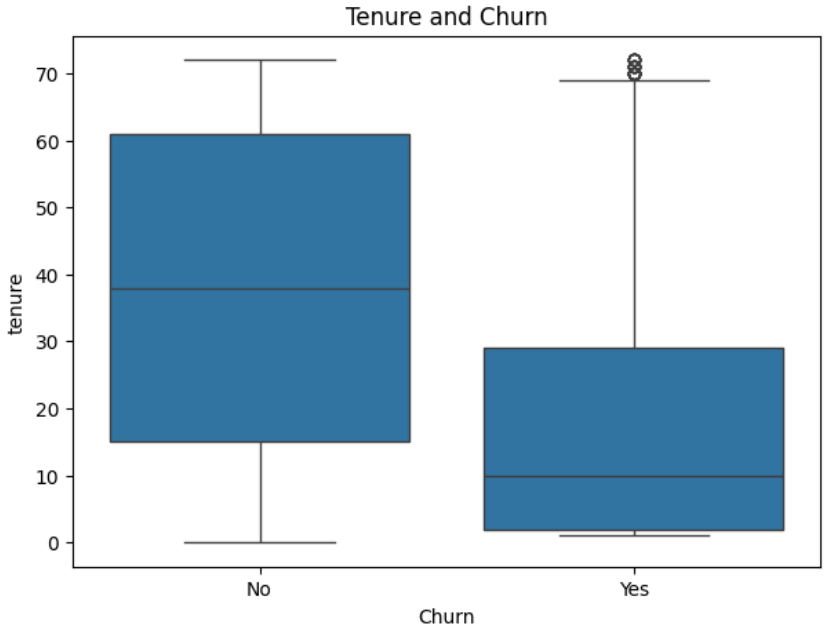
```
Q1: 402.225
Q3: 3786.6
IQR: 3384.375
Lower Bound: -4674.3375
Upper Bound: 8863.1625
Number of Outliers: 0

Feature: tenure
Q1: 9.0
Q3: 55.0
IQR: 46.0
Lower Bound: -60.0
Upper Bound: 124.0
Number of Outliers: 0
```




Internet Service and Churn
Churn
No
Yes
count
2000
1750
1500
1250
1000
750
500
250
0
DSL
Fiber optic
No
InternetService


Monthly Charges and Churn
MonthlyCharges
120
100
80
60
40
20
No
Yes
Churn




```
Churn Percentage by Contract Type:
Churn                  No        Yes
Contract
Month-to-month  57.290323  42.709677
One year        88.730482  11.269518
Two year        97.168142   2.831858

Contract Type with Highest Churn: Month-to-month

Payment Method Used by Churned Customers:
PaymentMethod
Electronic check             1071
Mailed check                  308
Bank transfer (automatic)     258
Credit card (automatic)       232
Name: count, dtype: int64

Most Frequently Used Payment Method Among Churned Customers: Electronic check

Average Tenure by Churn:
Churn
No     37.569965
Yes    17.979133
Name: tenure, dtype: float64

Customers with shorter tenure tend to churn more often.

Average Monthly Charges by Churn:
Churn
No     61.265124
Yes    74.441332
```

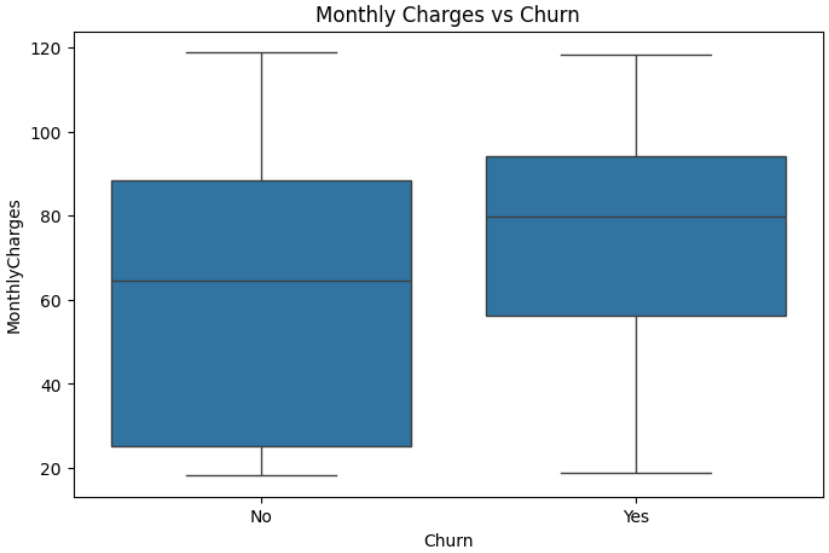
```
Name: MonthlyCharges, dtype: float64

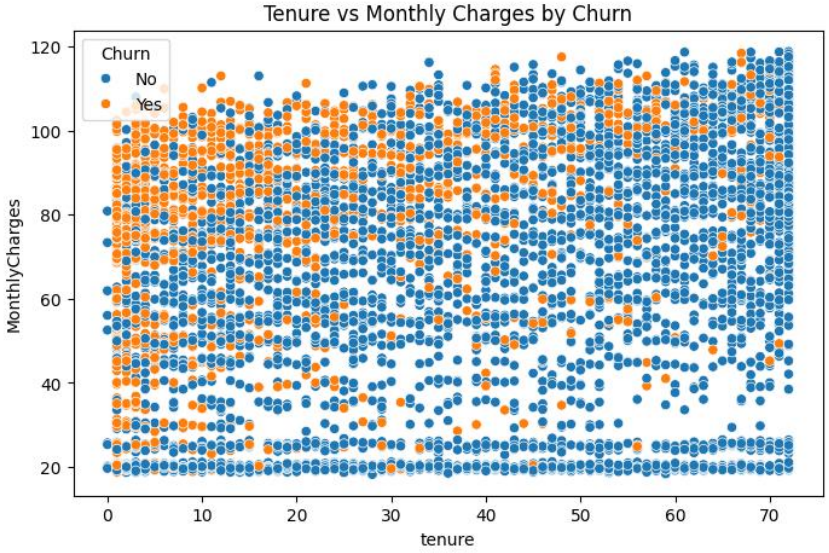
Customer Group with Highest Monthly Charges: Yes

Churn Percentage by Internet Service:
Churn                  No        Yes
InternetService
DSL              81.040892  18.959108
Fiber optic      58.107235  41.892765
No               92.595020   7.404980

Internet Service Category with Highest Churn: Fiber optic

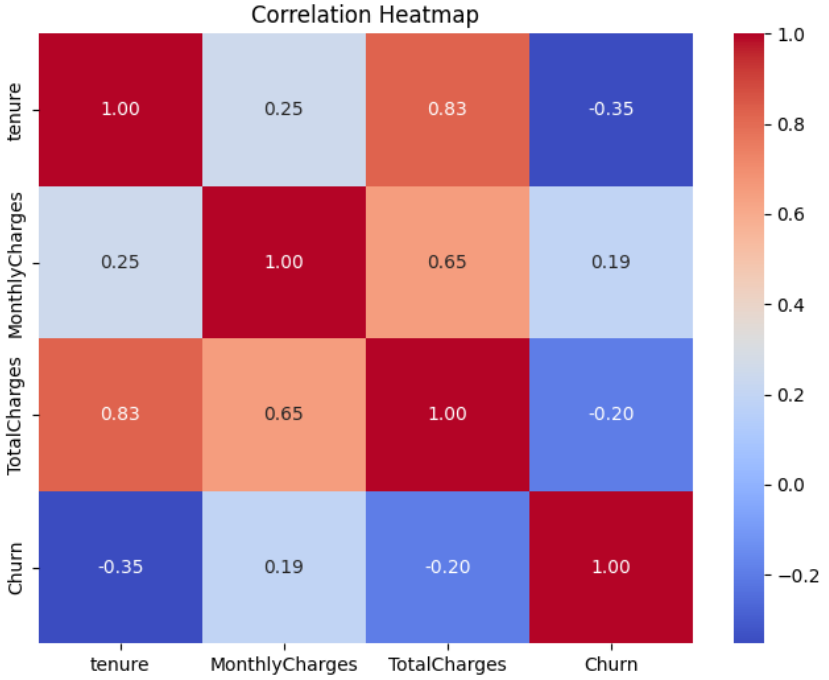
Correlation with Churn:
MonthlyCharges: 0.19335642223784694
Tenure: -0.352228670113078
TotalCharges: -0.1990368318692923
```





Correlation Matrix:

	tenure	MonthlyCharges	TotalCharges	Churn
tenure	1.000000	0.247900	0.825464	-0.352229
MonthlyCharges	0.247900	1.000000	0.650864	0.193356
TotalCharges	0.825464	0.650864	1.000000	-0.199037
Churn	-0.352229	0.193356	-0.199037	1.000000



Training Data Shape: (5634, 7072)

```
Testing Data Shape: (1409, 7072)
Downloading "CustomerChurn_Preprocessed.csv":
```